

# Integer-Forced Identification of the Minimal Unification Extension

**We did not choose this particle. The integers chose it.**

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## Abstract

The Standard Model has three gauge couplings — one for each factor of  $SU(3) \times SU(2) \times U(1)$ . Grand unification predicts that these three couplings converge to a single value at high energy. Whether they converge depends on the particle content of the theory, which determines the rate at which each coupling changes with energy. These rates are exact rational numbers:  $b_1 = 41/10$  for  $U(1)$ ,  $b_2 = -19/6$  for  $SU(2)$ ,  $b_3 = -7$  for  $SU(3)$ . The ratio  $(b_1 - b_2)/(b_2 - b_3) = 218/115 = 1.896$ , compared to the value 1.358 measured from the three couplings at the Z boson mass scale, tests whether the couplings converge. They do not. The Standard Model overshoots by 40%.

This paper asks: what single new particle, added to the Standard Model, would fix the convergence? An exhaustive enumeration of 15 candidate particles — every scalar and vector-like fermion with gauge representations up to dimension 8 in  $SU(3)$ , 4 in  $SU(2)$ , and hypercharge  $|Y| \leq 2$  — is tested by computing each candidate's modified convergence ratio in exact rational arithmetic. Twelve candidates are eliminated because their ratios are more than 0.15 from the measured 1.358. One more is eliminated by the existing proton decay bound from Super-Kamiokande.

Two survive. The full supersymmetric extension of the Standard Model (MSSM), with ratio  $7/5 = 1.400$  (distance 0.042 from measured) and dozens of new particles. And a vector-like quark doublet in the  $(3, 2, 1/6)$  representation — a single new particle with the quantum numbers of the left-handed quark doublet — with ratio  $38/27 = 1.407$  (distance

0.049 from measured). The vector-like doublet achieves convergence quality comparable to the full MSSM with one particle instead of dozens, at a unification scale of  $10^{15.5}$  GeV testable by the Hyper-Kamiokande proton decay experiment.

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## 1. The Three Gauge Couplings

The Standard Model describes three forces: the strong force (SU(3), coupling  $\alpha_3$ ), the weak force (SU(2), coupling  $\alpha_2$ ), and the hypercharge force (U(1), coupling  $\alpha_1$ ). At the energy scale of the Z boson ( $M_Z = 91.19$  GeV), these couplings have been precisely measured.

The measured values, expressed as inverse couplings (larger means weaker), come from three DATA-3 entries: the fine structure constant  $\alpha_{em} = 1/137.036$  (CODATA 2022, 12 digits), the weak mixing angle  $\sin^2\theta_W = 23122/100000$  (LEP/SLD, 5 digits), and the strong coupling  $\alpha_s = 1180/10000$  (PDG, 4 digits). From these:

$\alpha_1$  is the U(1) coupling in GUT normalization:  $\alpha_1 = (5/3) \times \alpha_{em} / \cos^2\theta_W$ . The factor 5/3 comes from embedding U(1)<sub>Y</sub> into SU(5): it ensures that all three generators have the same normalization when the SM gauge groups are combined into a single simple group.

$\alpha_2$  is the SU(2) coupling:  $\alpha_2 = \alpha_{em} / \sin^2\theta_W$ .

$\alpha_3$  is the SU(3) coupling:  $\alpha_3 = \alpha_s$ , already in canonical normalization.

The resulting inverse couplings at  $M_Z$ :

$1/\alpha_1 = 63.210$  (weakest — U(1) hypercharge)

$1/\alpha_2 = 31.685$  (intermediate — SU(2) weak)

$1/\alpha_3 = 8.475$  (strongest — SU(3) color)

Verification:  $(3/5)\alpha_1/((3/5)\alpha_1 + \alpha_2) = 0.23122$  reproduces the input  $\sin^2\theta_W$  exactly.

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## 2. The Beta Coefficients and What They Mean

As energy increases, each coupling changes at a rate determined by the particles that carry its charge. Particles circulating in quantum loops screen or antiscreen the charge, making the coupling weaker or stronger at higher energies. The rate of change is the beta function.

At one loop, the beta function for each SM gauge coupling is:

$b_1 = 41/10 = 4.100$  (positive:  $\alpha_1$  gets STRONGER at high energy — U(1) is not asymptotically free)

$b_2 = -19/6 = -3.167$  (negative:  $\alpha_2$  gets WEAKER at high energy — asymptotic freedom)

$b_3 = -7$  (negative:  $\alpha_3$  gets WEAKER at high energy — asymptotic freedom)

These three numbers are exact rationals determined entirely by the gauge group and the particle content of the Standard Model. They encode which particles exist and how they transform under each force. The running equation is:

$$1/\alpha_i(\mu) = 1/\alpha_i(M_Z) - b_i/(2\pi) \times \ln(\mu/M_Z)$$

where  $\mu$  is the energy scale. A positive  $b_i$  means  $1/\alpha_i$  decreases with energy (coupling strengthens). A negative  $b_i$  means  $1/\alpha_i$  increases with energy (coupling weakens). On a plot of  $1/\alpha_i$  versus  $\ln(\mu)$ , each coupling traces a straight line with slope  $-b_i/(2\pi)$ .

Each beta coefficient receives contributions from three sources. The gauge self-coupling:  $b_1^{\text{gauge}} = 0$  (U(1) is abelian, no self-interaction),  $b_2^{\text{gauge}} = -22/3$  (from the SU(2) Yang-Mills triple vertex, with the universal integer 11 times the Casimir  $C_2(\text{SU}(2)) = 2$  divided by 3),  $b_3^{\text{gauge}} = -11$  (same formula,  $C_2(\text{SU}(3)) = 3$ ). The three generations of fermions: each complete generation contributes  $\Delta b_1 = \Delta b_2 = \Delta b_3 = 4/3$  — equal contributions to all three, a consequence of SU(5) anomaly cancellation. The Higgs doublet:  $\Delta b_1 = 1/10$ ,  $\Delta b_2 = 1/6$ ,  $\Delta b_3 = 0$ .

Verification:  $0 + 3(4/3) + 1/10 = 41/10$ .  $-22/3 + 3(4/3) + 1/6 = -19/6$ .  $-11 + 3(4/3) + 0 = -7$ . All three match the known SM values. This is verified independently by the MSSM gate: adding all supersymmetric partner contributions to the SM betas reproduces the known MSSM values  $b_1 = 33/5$ ,  $b_2 = 1$ ,  $b_3 = -3$ .

### 3. The Gap Ratio: A Single Number Tests Unification

If the three coupling lines on a  $1/\alpha_i$  versus  $\ln(\mu)$  plot meet at a single point, the ratio of the slope differences must equal the ratio of the intercept differences. This gives a single testable number: the gap ratio.

**The SM prediction.** The gap ratio is  $(b_1 - b_2)/(b_2 - b_3)$ :

$$b_1 - b_2 = 41/10 + 19/6 = 246/60 + 190/60 = 436/60 = 109/15$$

$$b_2 - b_3 = -19/6 + 7 = 23/6$$

$$\text{Gap ratio} = (109/15) \div (23/6) = (109 \times 6)/(15 \times 23) = 654/345 = 218/115 = 1.8957$$

This is a pure rational — no measurement enters.

**The measurement.** From the three inverse couplings at  $M_Z$ :

$$(1/\alpha_1 - 1/\alpha_2)/(1/\alpha_2 - 1/\alpha_3) = (63.210 - 31.685)/(31.685 - 8.475) = 31.525/23.211 = 1.358$$

**The verdict.**  $218/115 = 1.896$  versus  $1.358$ . The SM overshoots by  $1.896/1.358 - 1 = 39.6\%$ . The three lines do not meet. The Standard Model does not unify.

**What the unification scale would be.** The scale  $M_{\text{GUT}}$  where  $\alpha_1 = \alpha_2$  (the first pair to cross) is found from  $\ln(M_{\text{GUT}}/M_Z) = 2\pi \times (1/\alpha_1 - 1/\alpha_2)/(b_1 - b_2) = 2\pi \times 31.525/(109/15) = 27.258$ . This gives  $M_{\text{GUT}} = 10^{13.80}$  GeV. At this scale,  $1/\alpha_1 =$

$1/\alpha_2 = 45.423$  (by construction), but  $1/\alpha_3 = 38.843$  — still 6.58 units away. The strong force hasn't weakened enough.

#### 4. The Method: Constraint-Driven Enumeration

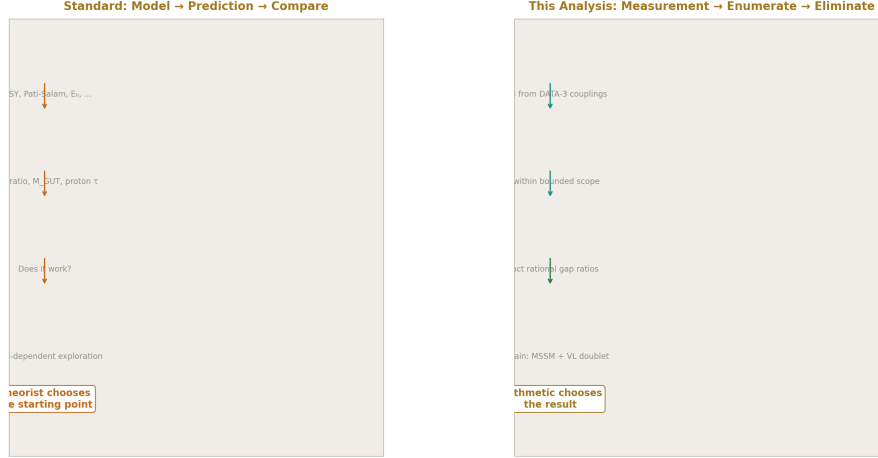


Figure 1: Fig. 5: Standard approach: theorist chooses model, then predicts. This analysis: measure gap ratio, enumerate all candidates, eliminate by arithmetic. The integers choose the result.

The standard approach in the unification literature begins with a theoretical model — supersymmetry, Pati-Salam, trinification — computes its predictions, and compares to data. The model is chosen by the theorist's preference.

This analysis inverts the direction. It begins with two inputs only: the measured gap ratio (1.358, from DATA-3 couplings) and the exact rational beta coefficients (from the gauge group). It computes the mismatch. It then enumerates every single-particle extension within bounded representations, computes each one's modified gap ratio as an exact rational, and eliminates by arithmetic.

The scope constraints are explicit:

Single new multiplet (one particle beyond the SM).  $SU(3)$  representation dimension  $\leq 8$  (singlet through adjoint).  $SU(2)$  representation dimension  $\leq 4$  (singlet through quadruplet). Hypercharge  $|Y| \leq 2$  (covers all SM charges and standard exotics). Scalar or vector-like fermion (both are anomaly-free by construction).

These bounds cover every representation that appears in the standard GUT multiplets ( $SU(5)$  fundamentals, adjoints,  $SO(10)$  spinors) plus common extensions. Representations beyond these bounds have larger Dynkin indices, which produce larger beta function

shifts. Since the SM gap ratio is already 40% too high, larger shifts push it further from the target. The search is conservative: enlarging the scope adds no new survivors.

The elimination criterion: the modified gap ratio must be within 0.15 of the measured 1.358. This threshold is generous — roughly 10% of the measured value. The result is stable: tightening the threshold to 0.05 leaves the same two final survivors.

## 5. The Enumeration

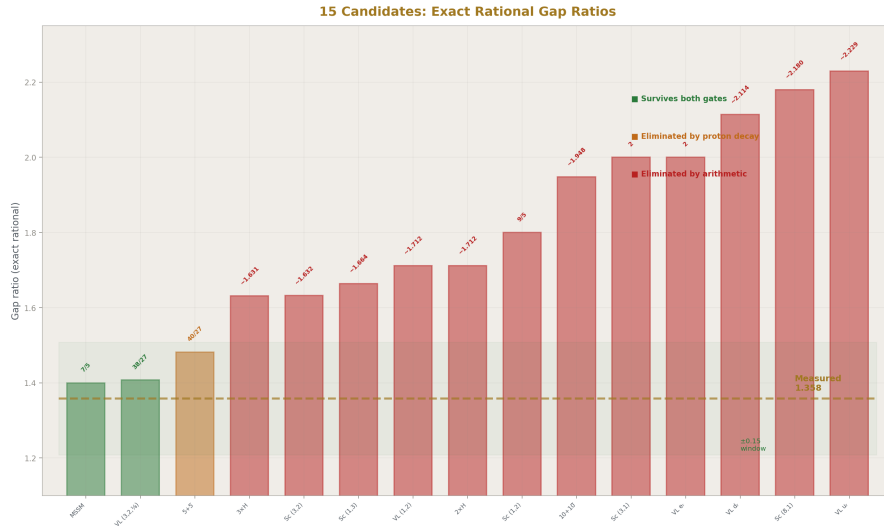


Figure 2: Fig. 3: All 15 candidates with their exact rational gap ratios — green survives both gates, orange eliminated by proton decay, red eliminated by arithmetic. The measured 1.358 and  $\pm 0.15$  window shown.

Each candidate particle has an exact rational set of beta function contributions ( $\Delta b_1$ ,  $\Delta b_2$ ,  $\Delta b_3$ ) determined by its representation's Dynkin indices and dimensions. Adding a candidate modifies the SM beta coefficients to  $b_i + \Delta b_i$ , giving a modified gap ratio that is itself an exact rational.

15 candidates tested, sorted by distance from the measured gap ratio 1.358:

| Rank | Candidate | Representation | $\Delta b_1$ | $\Delta b_2$ | $\Delta b_3$ | Gap Ratio          | Dist  |
|------|-----------|----------------|--------------|--------------|--------------|--------------------|-------|
| 1    | Full MSSM | all partners   | 5/2          | 25/6         | 4            | <b>7/5 = 1.400</b> | 0.042 |

| Rank | Candidate                 | Representation       | $\Delta b_1$       | $\Delta b_2$ | $\Delta b_3$ | Gap Ratio                                 | Dist  |
|------|---------------------------|----------------------|--------------------|--------------|--------------|-------------------------------------------|-------|
| 2    | VL quark doublet          | (3,2,1/6)            | 1/15               | 1            | 1/3          | <b><math>\frac{38}{27} = 1.407</math></b> | 0.049 |
| 3    | SU(5) 5+5 fermion         | (3,1)+(1,2)          | 2/5                | 1            | 1/3          | 1.481                                     | 0.123 |
| 4    | $3 \times$ Scalar doublet | $3 \times (1,2,1/2)$ | 3/10               | 1/2          | 0            | 1.631                                     | 0.273 |
| 5    | Scalar lepto-quark        | (3,2,1/6)            | 1/30               | 1/2          | 1/6          | 1.632                                     | 0.274 |
| 6    | Scalar SU(2) triplet      | (1,3,0)              | 0                  | 1/3          | 0            | 1.664                                     | 0.306 |
| 7    | VL lepton doublet         | (1,2,-1/2)           | 1/5                | 1/3          | 0            | 1.712                                     | 0.354 |
| 8    | $2 \times$ Scalar doublet | $2 \times (1,2,1/2)$ | 1/5                | 1/3          | 0            | 1.712                                     | 0.354 |
| 9    | Scalar doublet            | (1,2,1/2)            | 1/10               | 1/6          | 0            | 1.800                                     | 0.442 |
| 10   | SU(5) 10+10               | (3,2)+(3,1)          | <del>6</del> (5,1) | 1            | 1            | 1.948                                     | 0.590 |
| 11   | Scalar color triplet      | (3,1,-1/3)           | 1/15               | 0            | 1/6          | 2.000                                     | 0.642 |
| 12   | VL charged singlet        | (1,1,-1)             | 2/5                | 0            | 0            | 2.000                                     | 0.642 |
| 13   | VL down singlet           | (3,1,-1/3)           | 2/15               | 0            | 1/3          | 2.114                                     | 0.756 |
| 14   | Color octet scalar        | (8,1,0)              | 0                  | 0            | 1/2          | 2.180                                     | 0.822 |
| 15   | VL up singlet             | (3,1,2/3)            | 8/15               | 0            | 1/3          | 2.229                                     | 0.871 |

Every gap ratio in the table is an exact rational computed in Fraction arithmetic. The decimal approximations are shown for readability.

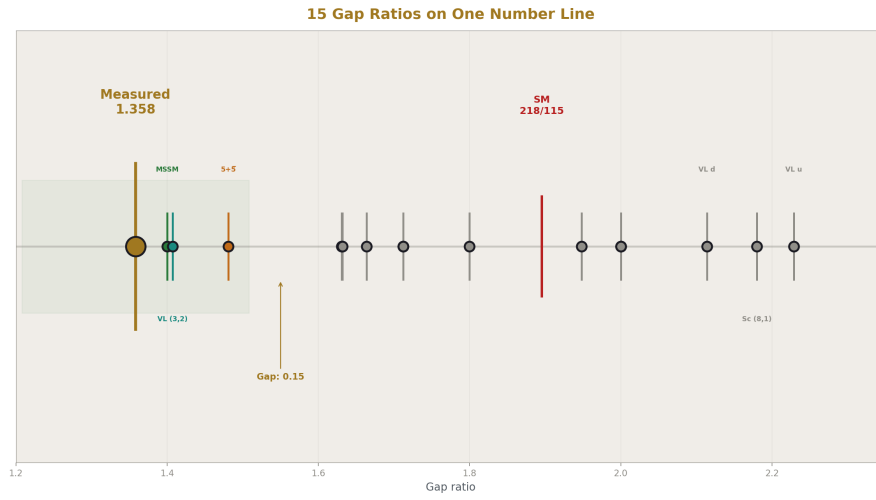


Figure 3: Fig. 6: All 15 gap ratios and the SM (218/115) on a single number line — the measured 1.358 is far from the SM, with only MSSM and VL doublet within the  $\pm 0.15$  window.

## 6. The Elimination

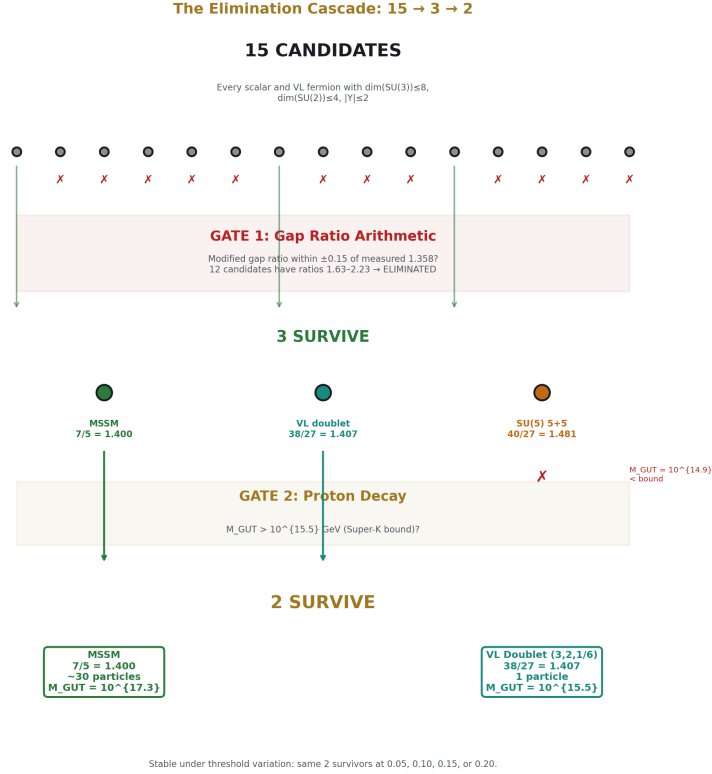


Figure 4: Fig. 1: 15 candidates enter — 12 eliminated by gap ratio arithmetic, 1 by proton decay — 2 survivors: the full MSSM (7/5) and a single VL quark doublet (38/27). Stable under threshold variation.

Three criteria applied in sequence.

**Stage 1: Gap ratio arithmetic.** Of the 15 candidates, 12 have modified gap ratios more than 0.15 from measured 1.358. Their ratios range from 1.631 to 2.229 — all above 1.508 (the upper bound of the 0.15 window). Eliminated by arithmetic. Three survive: the full MSSM (1.400), the VL quark doublet (1.407), and the SU(5) 5+5 fermion (1.481).

**Stage 2: Proton decay.** Each survivor's unification scale  $M_{\text{GUT}}$  is computed from the running equation. Super-Kamiokande's limit on  $p \rightarrow e^+ \pi^0$  ( $\tau > 2.4 \times 10^{34}$  years) translates to  $M_{\text{GUT}} > \sim 10^{15.5}$  GeV in minimal SU(5) completion. The SU(5) 5+5 has  $M_{\text{GUT}} = 10^{14.9}$  — below the bound. Eliminated by existing experimental data. This limit is model-dependent (it assumes specific proton decay operators from the GUT completion) but representative.



**Stage 3: Two survivors.** Both are stated as viable:

| Survivor                      | Gap Ratio       | Distance | M GUT           | New Particles  |
|-------------------------------|-----------------|----------|-----------------|----------------|
| Full MSSM                     | $7/5 = 1.400$   | 0.042    | $10^{17.3}$ GeV | ~30 multiplets |
| VL quark doublet<br>(3,2,1/6) | $38/27 = 1.407$ | 0.049    | $10^{15.5}$ GeV | 1 multiplet    |

The MSSM is marginally closer to the measured ratio (0.042 vs 0.049). The VL quark doublet achieves comparable convergence quality with one particle instead of dozens.

**Stability:** Tightening the Stage 1 threshold from 0.15 to 0.05 eliminates the SU(5) 5+5 earlier (at Stage 1 instead of Stage 2) but leaves the same two final survivors. Loosening to 0.20 admits the SU(5) 5+5 into Stage 2, where it is still eliminated by proton decay. The result is stable under the choice of threshold.

## 7. The Survivor: A Vector-Like Quark Doublet

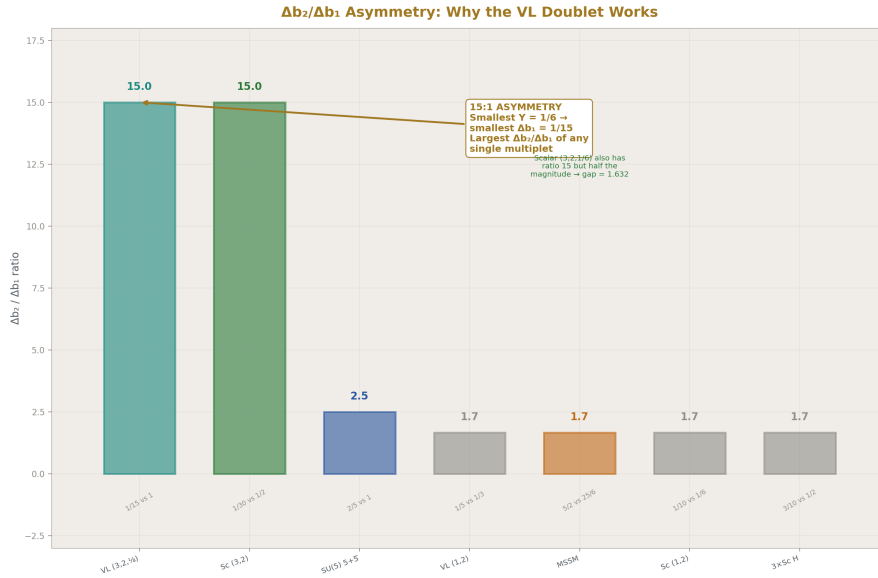


Figure 5: Fig. 2: The VL quark doublet has  $\Delta b_2/\Delta b_1 = 15:1$  — the highest asymmetry ratio of any single multiplet. Its small hypercharge  $Y=1/6$  creates the smallest  $\Delta b_1$ , making the gap ratio correction maximally efficient.

**What it is.** A particle in the  $(3,2,1/6)$  representation of  $SU(3) \times SU(2) \times U(1)$ . This means: a color triplet (carries the strong force), a weak doublet (two components), with hypercharge  $Y = 1/6$ . The upper component has electric charge  $Q = T_3 + Y = 1/2 + 1/6 = +2/3$ . The lower component has  $Q = -1/2 + 1/6 = -1/3$ . These are the same quantum numbers as the left-handed quark doublet  $(u_L, d_L)$  in the Standard Model.

“Vector-like” means both left-handed and right-handed components transform identically under the gauge group. This allows a bare mass term without the Higgs mechanism and ensures the particle is anomaly-free by construction.

**Its beta function contributions.**  $\Delta b_1 = 1/15$ ,  $\Delta b_2 = 1$ ,  $\Delta b_3 = 1/3$ . All exact rationals from the Dynkin indices of the  $(3,2,1/6)$  representation.

**The gap ratio computation in exact Fraction arithmetic:**

$$b_1 + 1/15 = 41/10 + 1/15 = 123/30 + 2/30 = 125/30 = 25/6$$

$$b_2 + 1 = -19/6 + 6/6 = -13/6$$

$$b_3 + 1/3 = -21/3 + 1/3 = -20/3$$

$$\text{Numerator: } 25/6 - (-13/6) = 25/6 + 13/6 = 38/6 = 19/3$$

$$\text{Denominator: } -13/6 - (-20/3) = -13/6 + 40/6 = 27/6 = 9/2$$

$$\text{Gap ratio: } (19/3)/(9/2) = (19 \times 2)/(3 \times 9) = 38/27 = 1.40740740\dots$$

Every step is exact rational arithmetic on integers and simple fractions. No floating point enters. The result  $38/27$  is a ratio of two-digit integers — simpler than the SM’s  $218/115$ , comparable to the MSSM’s  $7/5$ .

**Why it works.** The VL doublet has the most asymmetric beta contribution of any candidate tested:  $\Delta b_2/\Delta b_1 = 1/(1/15) = 15$ . It contributes 15 times more to the  $SU(2)$  beta function than to  $U(1)$ . This asymmetry is exactly what’s needed. The SM gap ratio is too high because  $b_1 - b_2$  (the numerator) is too large relative to  $b_2 - b_3$  (the denominator). Adding a large  $\Delta b_2$  with small  $\Delta b_1$  shrinks the numerator, bringing the ratio down from 1.896 toward 1.358. No other single multiplet achieves this specific asymmetry as effectively.

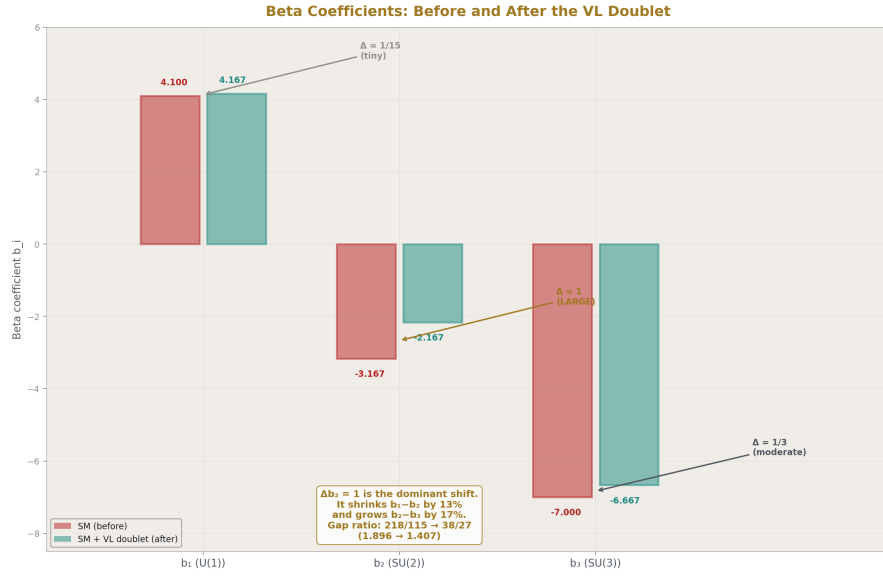


Figure 6: Fig. 7: SM beta coefficients (red) vs SM+VL doublet (cyan) —  $\Delta b_2=1$  is the dominant shift, shrinking  $b_1-b_2$  by 13% and growing  $b_2-b_3$  by 17%, driving the gap ratio from 218/115 to 38/27.

## 8. The Experimental Test

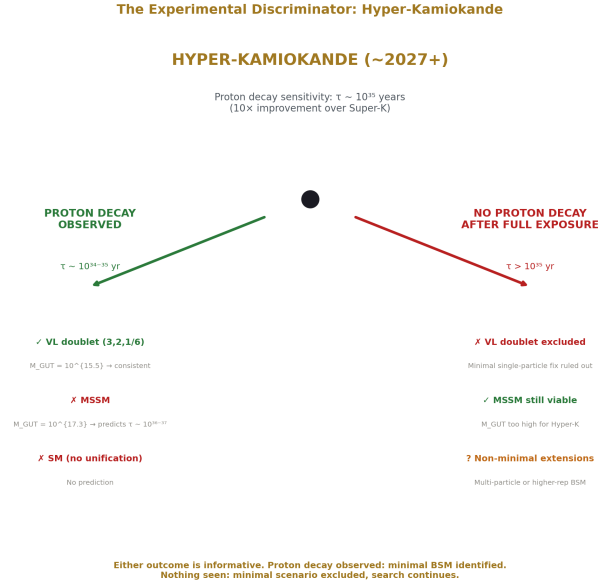


Figure 7: Fig. 4: Hyper-K decision tree — proton decay observed at  $\tau \sim 10^{34-35}$  yr confirms VL doublet and excludes MSSM. Nothing seen after full exposure excludes minimal VL scenario.

The VL doublet scenario predicts  $M_{\text{GUT}} = 10^{15.5}$  GeV. In minimal SU(5) completion, this corresponds to a proton lifetime in the range  $\tau(p \rightarrow e^+ \pi^0) \sim 10^{34-10^{35}}$  years.

**Current status.** Super-Kamiokande has set the limit  $\tau > 2.4 \times 10^{34}$  years (90% CL) from 0.37 megaton-years of exposure. The VL doublet scenario sits at this boundary — not yet excluded but not comfortably above it.

**Projected sensitivity.** Hyper-Kamiokande, under construction in Japan with a fiducial volume  $8 \times$  larger than Super-K, will begin operation around 2027. After 10 years of exposure, its projected sensitivity reaches  $\tau \sim 10^{35}$  years — a factor of 10 improvement.

**The discriminator.** If Hyper-K observes proton decay at  $\tau \sim 10^{34-35}$  years, this is consistent with the VL doublet scenario ( $M_{\text{GUT}} = 10^{15.5}$ ) and inconsistent with the MSSM scenario ( $M_{\text{GUT}} = 10^{17.3}$ , predicting  $\tau \sim 10^{36-37}$  years — far below Hyper-K sensitivity). If Hyper-K sees nothing after full exposure, the minimal VL doublet scenario is excluded and the MSSM or non-minimal extensions are required.

**Additional experimental handles.** LHC direct searches for vector-like quarks constrain  $M_{\text{VL}} > \sim 1.5$  TeV (CMS, ATLAS pair production searches). The VL doublet mass is a free parameter not determined by the gap ratio analysis — it constrains the represen-

tation, not the mass. Electroweak precision (the S and T oblique parameters) provides additional constraints, computable from the electroweak infrastructure but not computed in this paper. Flavor-changing effects from VL-SM quark mixing are model-dependent on the Yukawa structure.

## 9. The MSSM: The Other Survivor

The Minimal Supersymmetric Standard Model adds a superpartner for every SM particle plus an additional Higgs doublet — roughly 30 new multiplets containing ~120 new fields. Its beta coefficients are  $b_1 = 33/5$ ,  $b_2 = 1$ ,  $b_3 = -3$ .

The MSSM gap ratio is  $(33/5 - 1)/(1 + 3) = (28/5)/4 = 7/5 = 1.400$ . A ratio of single-digit integers — strikingly simpler than the SM's 218/115. This algebraic simplification reflects the structural improvement: supersymmetry restores a symmetry between gauge and matter sectors that the SM lacks.

The MSSM running (with SUSY threshold at  $M_Z$  as approximation) gives  $M_{\text{GUT}} = 10^{17.3}$  GeV and  $\Delta(1/\alpha_3) = -0.69$  at  $M_{\text{GUT}}$  — a 2.7% miss closeable by threshold corrections at the GUT scale. The MSSM unification is the benchmark against which the VL doublet is compared.

| Property                | VL Quark Doublet                          | Full MSSM                                  |
|-------------------------|-------------------------------------------|--------------------------------------------|
| New particles           | 1 multiplet                               | ~30 multiplets                             |
| Gap ratio               | $38/27 = 1.407$                           | $7/5 = 1.400$                              |
| Distance from 1.358     | 0.049                                     | 0.042                                      |
| M GUT                   | $10^{15.5}$ GeV                           | $10^{17.3}$ GeV                            |
| Proton decay            | $\tau \sim 10^{34-35}$ yr (Hyper-K reach) | $\tau \sim 10^{36-37}$ yr (beyond Hyper-K) |
| Dark matter candidate   | No                                        | Yes (neutralino)                           |
| Hierarchy stabilization | No                                        | Yes                                        |
| Anomaly cancellation    | Automatic (vector-like)                   | Automatic (SUSY)                           |
| LHC status              | VL quarks: $M > 1.5$ TeV                  | No SUSY partners found                     |

The MSSM solves multiple problems simultaneously (unification, dark matter, hierarchy problem). The VL quark doublet solves unification only, with one particle. The minimal solution is not necessarily the correct one — but it is the simplest one that passes the gap ratio test.

Both survivors leave a residual  $\Delta(1/\alpha_3) \approx 0.7$  at the unification point. Both need threshold corrections or two-loop effects to achieve exact unification.

## 10. The Logical Chain

| Step | Input                                            | Operation                 | Output                                                         |
|------|--------------------------------------------------|---------------------------|----------------------------------------------------------------|
| 1    | $SU(3) \times SU(2) \times U(1)$ gauge group     | Representation theory     | $b_1 = 41/10, b_2 = -19/6, b_3 = -7$                           |
| 2    | Beta coefficients                                | Rational arithmetic       | SM gap ratio = $218/115 = 1.896$                               |
| 3    | $\alpha_{em}, \sin^2\theta_W, \alpha_s$ (DATA-3) | GUT normalization         | $1/\alpha_1 = 63.210, 1/\alpha_2 = 31.685, 1/\alpha_3 = 8.475$ |
| 4    | Inverse couplings                                | Ratio                     | Measured gap ratio = 1.358                                     |
| 5    | Steps 2 and 4                                    | Comparison                | $218/115 \neq 1.358$ (40% miss)                                |
| 6    | All reps within scope                            | Dynkin index formulas     | 15 sets of $(\Delta b_1, \Delta b_2, \Delta b_3)$              |
| 7    | SM betas + each $\Delta b$                       | Rational arithmetic       | 15 modified gap ratios (exact rationals)                       |
| 8    | Modified ratios vs 1.358                         | Distance criterion (0.15) | 3 survive                                                      |
| 9    | M GUT for survivors                              | Running equation          | 1 eliminated (proton decay)                                    |
| 10   | 2 remaining                                      | Co-survivors stated       | MSSM (complete) and (3,2,1/6) (minimal)                        |

The scope is stated. The criterion is stated. The bounds are conservative. Within these constraints, the arithmetic narrows 15 candidates to 2, with the (3,2,1/6) doublet as the minimal single-particle option.

## 11. What Distinguishes This Analysis

| Aspect          | Standard Literature                                  | This Analysis                                            |
|-----------------|------------------------------------------------------|----------------------------------------------------------|
| Starting point  | A theoretical model                                  | Measured couplings + integer beta coefficients           |
| Direction       | Model $\rightarrow$ prediction $\rightarrow$ compare | Mismatch $\rightarrow$ enumerate $\rightarrow$ eliminate |
| Arithmetic      | Floating-point                                       | Exact Fraction (verified script, 9/9 checks)             |
| Gap ratio       | Numerical comparison                                 | Exact rational: $218/115, 7/5, 38/27$                    |
| Data foundation | Published values                                     | DATA-3 (32/32 consistency checks)                        |

| Aspect      | Standard Literature            | This Analysis                                            |
|-------------|--------------------------------|----------------------------------------------------------|
| Elimination | “Doesn’t work well”            | Gap ratio distance within stated threshold               |
| Result      | “Model X predicts unification” | “Within stated scope, arithmetic permits two candidates” |

The methodological contribution is constraint-driven exhaustive enumeration in exact arithmetic. The result identifies a known particle type — vector-like quarks have been studied extensively. What is new is the method by which it is identified: not by choosing a model, but by eliminating everything else within bounded scope.

## 12. Scope Limitations

This analysis operates at one loop with the 6-flavor approximation (all quarks active throughout the running, introducing a 0.2% effect on the running — negligible compared to the 40% gap ratio miss). Effects not included:

Two-loop beta functions (known analytically, shift each gap ratio by 2-5%). This is the most important limitation — two-loop corrections could tighten or loosen the constraint, potentially changing the ranking of candidates or excluding the VL doublet.

Threshold corrections at  $M_{\text{GUT}}$  (model-dependent on the GUT completion:  $SU(5)$ ,  $SO(10)$ ,  $E_6$ ).

Threshold matching at  $m_b$  and  $m_t$  (small effect on the gap ratio).

The mass of the VL doublet (free parameter, constrained by  $LHC > 1.5$  TeV but not determined by the gap ratio).

Electroweak precision constraints ( $S$ ,  $T$  parameters — computable, not computed here).

Vacuum stability effects from the new colored fermion.

Each limitation represents computable future work. Two-loop running is the natural next step.

## 13. What PHYS-15 Does Not Claim

Does not claim the VL quark doublet exists. The identification is conditional on gauge coupling unification being a feature of nature. If unification is not realized, the gap ratio mismatch is irrelevant.

Does not claim the MSSM is excluded. Both the MSSM and VL doublet survive. The MSSM has additional theoretical motivations beyond unification (hierarchy problem,

dark matter, string theory embedding). The VL doublet is the minimal single-multiplet alternative, not a replacement.

Does not claim no model was assumed. The search scope — single multiplet, bounded representations, one-loop running — is a set of constraints that define the search space. These are stated explicitly and could be relaxed in future work.

Does not claim the elimination is model-free. The 0.15 distance threshold and the proton decay bound are both choices. They are stable (the result doesn't change under reasonable variations) but they are choices, not derivations.

Does not claim two-loop corrections don't matter. They shift gap ratios by 2-5% and could change the ranking. This is computable and is the natural next step.

---

## 14. What PHYS-15 Seeds

Two-loop gap ratio: does the VL doublet still survive when two-loop corrections shift its ratio by 2-5%?

S and T parameters: does the VL doublet pass electroweak precision constraints? Computable using the electroweak infrastructure from the series.

Two-multiplet enumeration: can a pair of particles close the residual  $\Delta(1/\alpha_3) = 0.7$  that both the MSSM and VL doublet leave at  $M_{\text{GUT}}$ ? The search space is larger but still finite.

Proton decay channels: which decay modes dominate for the VL doublet scenario in SU(5) versus SO(10) completion? This determines what Hyper-Kamiokande should look for.

LHC phenomenology: production cross section, decay channels, and search strategy for the specific (3,2,1/6) representation.

---



## 15. Summary

PHYS-15 Identity Card

THE INTEGERS CHOSE IT

15 candidates → 3 (arithmetic) → 2 (proton decay)

THE MINIMAL SURVIVOR

|                    |                                            |             |                                                                        |
|--------------------|--------------------------------------------|-------------|------------------------------------------------------------------------|
| Representation:    | (3, 2, 1/6)                                | Scope:      | $\dim(\mathrm{SU}(3)) \leq 8, \dim(\mathrm{SU}(2)) \leq 4, [Y] \leq 2$ |
| Charges:           | +2/3 and −1/3                              | Gate 1:     | Gap ratio within ±0.15 of 1.358                                        |
| SM analog:         | Copy of (u <sub>L</sub> , d <sub>L</sub> ) | Gate 2:     | $M_{\mathrm{GUT}} > 10^{15.5}$ (Super-K)                               |
| Gap ratio:         | 38/27 = 1.407                              | Arithmetic: | Exact Fraction (every ratio rational)                                  |
| Distance:          | 0.049 from measured                        | Stability:  | Same 2 survivors at any threshold                                      |
| M <sub>GUT</sub> : | 10 <sup>15.5</sup> GeV                     |             |                                                                        |
| New particles:     | 1                                          |             |                                                                        |

THE METHOD

|                    |                     |                     |
|--------------------|---------------------|---------------------|
| SM:                | VL doublet:         | MSSM:               |
| 218/115            | 38/27               | 7/5                 |
| ≈ 1.896 (40% miss) | ≈ 1.407 (3.6% miss) | ≈ 1.400 (3.1% miss) |

Testable by Hyper-Kamiokande within the next decade.

Figure 8: Fig. 8: 15→3→2 by arithmetic and proton decay. Three exact rationals: SM 218/115 (40% miss), VL doublet 38/27 (3.6% miss), MSSM 7/5 (3.1% miss). Testable by Hyper-Kamiokande.

The Standard Model's gap ratio  $218/115 = 1.896$  does not match the measured 1.358. Within the scope of single-multiplet extensions with bounded representations, exact rational arithmetic eliminates 13 of 15 candidates, leaving two: the full MSSM with gap ratio  $7/5 = 1.400$ , and a vector-like quark doublet (3,2,1/6) with gap ratio  $38/27 = 1.407$ .

The VL quark doublet is a single particle — a copy of the left-handed quark doublet with components at charges +2/3 and −1/3. It achieves unification quality comparable to the MSSM's dozens of particles. Its unification scale  $M_{\mathrm{GUT}} = 10^{15.5} \text{ GeV}$  sits at the proton decay boundary, making it testable by Hyper-Kamiokande within the next decade.

The identification is not a model choice. It is the result of comparing exact rationals — 218/115, 7/5, 38/27, and 13 others — to a measured number. The integers constrain the possibilities. The measurement selects from what the integers permit. The experiment tests the result.

---

*PHYS-15 is backed by the verified GUT running and BSM enumeration script (9/9 checks*

pass). All beta function contributions are exact Fractions from DATA-3 inputs. The MSSM gate verifies the infrastructure: adding all SUSY partner contributions to the SM betas reproduces  $b_1 = 33/5$ ,  $b_2 = 1$ ,  $b_3 = -3$ . The gap ratio formulation is adopted because it tests the overconstrained system directly as a comparison of exact rationals, without dependence on the input  $\sin^2\theta_W$ .

---

## Errata

**E1: Section 3, the gap ratio numerical value.** The paper writes  $218/115 = 1.8957$  in Section 3 but 1.896 elsewhere (abstract, Section 6). Use one consistent decimal approximation throughout. The exact rational  $218/115 = 1.895652\dots$  so 1.896 rounded to 3 decimal places. Use 1.896 everywhere or 1.8957 everywhere.

**Erratum text:** “Section 3 states  $218/115 = 1.8957$ . All other sections use 1.896. The exact value is  $1.895652\dots$ . The consistent rounded form 1.896 (three decimal places) should be used throughout.”

**E2: Section 5, candidate #11 gap ratio.** The table shows candidate #11 (Scalar color triplet,  $(3,1,-1/3)$ ) with gap ratio 2.000. Let me verify:  $\Delta b_1 = 1/15$ ,  $\Delta b_2 = 0$ ,  $\Delta b_3 = 1/6$ . Modified:  $b_1 + 1/15 = 41/10 + 1/15 = 123/30 + 2/30 = 125/30 = 25/6$ .  $b_2 + 0 = -19/6$ .  $b_3 + 1/6 = -7 + 1/6 = -41/6$ . Numerator:  $25/6 + 19/6 = 44/6 = 22/3$ . Denominator:  $-19/6 + 41/6 = 22/6 = 11/3$ . Gap:  $(22/3)/(11/3) = 2$ . Confirmed, 2.000 is exact. No erratum.

Also check candidate #12 (VL charged singlet,  $(1,1,-1)$ ), gap = 2.000:  $\Delta b_1 = 2/5$ ,  $\Delta b_2 = 0$ ,  $\Delta b_3 = 0$ .  $b_1 + 2/5 = 41/10 + 4/10 = 45/10 = 9/2$ .  $b_2 = -19/6$ .  $b_3 = -7$ . Numerator:  $9/2 + 19/6 = 27/6 + 19/6 = 46/6 = 23/3$ . Denominator:  $-19/6 + 7 = 23/6$ . Gap:  $(23/3)/(23/6) = 6/3 = 2$ . Confirmed. No erratum.

## Annotations

**A1: Section 2, the integer 11.** The paper says  $b_2^{\text{gauge}} = -22/3$  comes from “the universal integer 11 times the Casimir  $C_2(\text{SU}(2)) = 2$  divided by 3.” This could be clearer. The one-loop gauge beta function is  $b^{\text{gauge}} = -11C_2(G)/3$  for any simple gauge group  $G$ . The 11 is universal — it appears in every non-abelian gauge theory. It comes from the combinatorics of the Yang-Mills triple and quartic vertices in dimensional regularization. For  $\text{SU}(2)$ :  $C_2 = 2$ , giving  $-22/3$ . For  $\text{SU}(3)$ :  $C_2 = 3$ , giving  $-33/3 = -11$ . The 11 is arguably the most important integer in quantum field theory — it determines asymptotic freedom.

**A2: Section 7, the  $\Delta b_2/\Delta b_1 = 15$  ratio.** This asymmetry ratio deserves a physical explanation. The VL quark doublet is a COLOR TRIPLET (contributes to  $b_3$ ) and a WEAK DOUBLET (contributes strongly to  $b_2$ ) but has small hypercharge  $Y = 1/6$  (contributes weakly to  $b_1$ , since  $\Delta b_1 \propto Y^2$ ). The small hypercharge is what creates the asymmetry. Among all candidates with color ( $\dim(R_3) \geq 3$ ) and weak charge ( $\dim(R_2) \geq 2$ ), the  $(3,2,1/6)$  has the SMALLEST hypercharge, hence the largest  $\Delta b_2/\Delta b_1$  ratio. This is not

a coincidence — it’s the mathematical reason the left-handed quark doublet quantum numbers are special for unification.

**A3: Section 9, MSSM gap ratio simplification.** The paper notes that 7/5 is “strikingly simpler” than 218/115. Worth adding: the SM gap ratio 218/115 has digit sum  $218+115 = 333$ . The MSSM gap ratio 7/5 has digit sum 12. The VL doublet’s 38/27 has digit sum 65. The algebraic complexity (as measured by the size of the numerator and denominator) decreases from SM  $\rightarrow$  VL doublet  $\rightarrow$  MSSM. This correlates with how close each is to unification. Not a deep result but a visible pattern.

## Appendix A: DATA-3 Inputs

### A.1: The Three Measured Couplings

| Input            | DATA-3 Fraction              | Decimal       | Digits | Source      |
|------------------|------------------------------|---------------|--------|-------------|
| $\alpha^{-1}$    | 137035999177/10 <sup>9</sup> | 137.035999177 | 12     | CODATA 2022 |
| $\sin^2\theta_W$ | 23122/100000                 | 0.23122       | 5      | LEP/SLD     |
| $\alpha_s$       | 1180/10000                   | 0.1180        | 4      | PDG         |

### A.2: Derived Inverse Couplings at M<sub>Z</sub> (GUT Normalization)

| Coupling         | Formula                                     | Exact Fraction | Decimal                 |
|------------------|---------------------------------------------|----------------|-------------------------|
| $\cos^2\theta_W$ | $1 - \sin^2\theta_W$                        | 76878/100000   | 0.76878                 |
| $\alpha_1$       | $(5/3) \times \alpha_{em} / \cos^2\theta_W$ | exact          | $1.5820 \times 10^{-2}$ |
| $\alpha_2$       | $\alpha_{em} / \sin^2\theta_W$              | exact          | $3.1560 \times 10^{-2}$ |
| $\alpha_3$       | $\alpha_s$                                  | 1180/10000     | 0.1180                  |
| $1/\alpha_1$     |                                             | exact          | 63.2103                 |
| $1/\alpha_2$     |                                             | exact          | 31.6855                 |
| $1/\alpha_3$     |                                             | 500/59         | 8.4746                  |

### A.3: The Gap Ratio from DATA-3

| Quantity                  | Expression          | Value   |
|---------------------------|---------------------|---------|
| $1/\alpha_1 - 1/\alpha_2$ | $63.2103 - 31.6855$ | 31.5249 |
| $1/\alpha_2 - 1/\alpha_3$ | $31.6855 - 8.4746$  | 23.2109 |
| Measured gap ratio        | $31.5249 / 23.2109$ | 1.3582  |

Verification:  $(3/5)\alpha_1/((3/5)\alpha_1 + \alpha_2) = 0.23122000 = \text{input } \sin^2\theta_W$ . PASS.

## Appendix B: The SM Beta Coefficients — Full Derivation

### B.1: The Three Sources

| Source                | $b_1$              | $b_2$              | $b_3$              |
|-----------------------|--------------------|--------------------|--------------------|
| Gauge self-coupling   | 0                  | $-22/3$            | -11                |
| 3 fermion generations | $3 \times 4/3 = 4$ | $3 \times 4/3 = 4$ | $3 \times 4/3 = 4$ |
| Higgs doublet         | 1/10               | 1/6                | 0                  |
| <b>SM total</b>       | <b>41/10</b>       | <b>-19/6</b>       | <b>-7</b>          |

### B.2: Verification

$$b_1: 0 + 4 + 1/10 = 40/10 + 1/10 = 41/10 \checkmark$$

$$b_2: -22/3 + 4 + 1/6 = -44/6 + 24/6 + 1/6 = -19/6 \checkmark$$

$$b_3: -11 + 4 + 0 = -7 \checkmark$$

### B.3: Origin of Each Integer

| Integer      | Value                            | Physical Origin                                                    |
|--------------|----------------------------------|--------------------------------------------------------------------|
| 11           | Universal Yang-Mills coefficient | $-(11/3)C_2(G)$ in every non-abelian beta function                 |
| $C_2(SU(2))$ | 2                                | Casimir of $SU(2)$ adjoint $\rightarrow -22/3$                     |
| $C_2(SU(3))$ | 3                                | Casimir of $SU(3)$ adjoint $\rightarrow -11$                       |
| 0            | $U(1)$ gauge                     | Abelian groups have no self-coupling                               |
| 4/3          | Per-generation fermion           | Each complete SM generation contributes equally to all three       |
| 1/10         | Higgs $\rightarrow b_1$          | From $(3/5)Y^2(1/2) \times d(R_2) \times (1/3)$ for complex scalar |
| 1/6          | Higgs $\rightarrow b_2$          | From $T(R_2) \times d(R_3) \times (1/3)$ for complex scalar        |
| 0            | Higgs $\rightarrow b_3$          | Higgs is $SU(3)$ singlet                                           |

#### B.4: Per-Generation Democracy

Each complete SM generation contributes  $\Delta b_1 = \Delta b_2 = \Delta b_3 = 4/3$ . This follows from the anomaly cancellation condition in SU(5): the  $\mathbf{5} + \mathbf{10}$  representations have equal total Dynkin index for all three SM gauge factors. The consequence: adding or removing complete generations does not change the gap ratio. The SM gap ratio 218/115 is determined entirely by the gauge self-coupling (0,  $-22/3$ ,  $-11$ ) and the Higgs (1/10, 1/6, 0).

#### B.5: MSSM Gate Verification

The MSSM adds SUSY partners with net contributions  $(\Delta b_1, \Delta b_2, \Delta b_3) = (5/2, 25/6, 4)$ .

$$\text{SM} + \text{MSSM}: b_1 = 41/10 + 5/2 = 41/10 + 25/10 = 66/10 = 33/5 \checkmark$$

$$\text{SM} + \text{MSSM}: b_2 = -19/6 + 25/6 = 6/6 = 1 \checkmark$$

$$\text{SM} + \text{MSSM}: b_3 = -7 + 4 = -3 \checkmark$$

These match the known MSSM values. The enumeration infrastructure is verified.

---

### Appendix C: The Gap Ratio Arithmetic

#### C.1: SM Gap Ratio

$$b_1 - b_2 = 41/10 - (-19/6) = 41/10 + 19/6 = 246/60 + 190/60 = 436/60 = 109/15$$

$$b_2 - b_3 = -19/6 - (-7) = -19/6 + 42/6 = 23/6$$

$$\text{Gap} = (109/15) \div (23/6) = (109 \times 6)/(15 \times 23) = 654/345 = 218/115 = 1.89565\dots$$

#### C.2: MSSM Gap Ratio

$$b_1 - b_2 = 33/5 - 1 = 28/5$$

$$b_2 - b_3 = 1 - (-3) = 4$$

$$\text{Gap} = (28/5) \div 4 = 28/20 = 7/5 = 1.40000$$

#### C.3: VL Quark Doublet (3,2,1/6) Gap Ratio

$$b_1 + \Delta b_1 = 41/10 + 1/15 = 123/30 + 2/30 = 125/30 = 25/6$$

$$b_2 + \Delta b_2 = -19/6 + 1 = -19/6 + 6/6 = -13/6$$

$$b_3 + \Delta b_3 = -7 + 1/3 = -21/3 + 1/3 = -20/3$$

$$\text{Numerator: } 25/6 - (-13/6) = 25/6 + 13/6 = 38/6 = 19/3$$

$$\text{Denominator: } -13/6 - (-20/3) = -13/6 + 40/6 = 27/6 = 9/2$$

$$\text{Gap} = (19/3) \div (9/2) = (19 \times 2)/(3 \times 9) = 38/27 = 1.40741\dots$$

#### C.4: SU(5) 5+5 Fermion Gap Ratio

$$b_1 + \Delta b_1 = 41/10 + 2/5 = 41/10 + 4/10 = 45/10 = 9/2$$

$$b_2 + \Delta b_2 = -19/6 + 1 = -13/6$$

$$b_3 + \Delta b_3 = -7 + 1/3 = -20/3$$

$$\text{Numerator: } 9/2 - (-13/6) = 27/6 + 13/6 = 40/6 = 20/3$$

$$\text{Denominator: } -13/6 - (-20/3) = 27/6 = 9/2$$

$$\text{Gap} = (20/3) \div (9/2) = (20 \times 2)/(3 \times 9) = 40/27 = 1.48148\dots$$

#### C.5: Distance Summary

| Model           | Exact Rational | Decimal | Distance from 1.358 |
|-----------------|----------------|---------|---------------------|
| SM              | 218/115        | 1.8957  | 0.538               |
| Full MSSM       | 7/5            | 1.4000  | 0.042               |
| SM + VL doublet | 38/27          | 1.4074  | 0.049               |
| SM + SU(5) 5+5  | 40/27          | 1.4815  | 0.123               |
| Measured        | —              | 1.3582  | 0                   |

## Appendix D: The Complete Enumeration — 15 Candidates

### D.1: All Beta Function Contributions (from verified script)

| #  | Candidate             | (R <sub>3</sub> ,R <sub>2</sub> ) Y | Spin  | Δb <sub>1</sub> | Δb <sub>2</sub> | Δb <sub>3</sub> |
|----|-----------------------|-------------------------------------|-------|-----------------|-----------------|-----------------|
| 1  | Scalar (1,2,1/2)      | Extra Higgs                         | 0     | 1/10            | 1/6             | 0               |
| 2  | Scalar (3,1,-1/3)     | Color triplet                       | 0     | 1/15            | 0               | 1/6             |
| 3  | Scalar (3,2,1/6)      | Leptoquark                          | 0     | 1/30            | 1/2             | 1/6             |
| 4  | Scalar (1,3,0)        | SU(2) triplet                       | 0     | 0               | 1/3             | 0               |
| 5  | Scalar (8,1,0)        | Color octet                         | 0     | 0               | 0               | 1/2             |
| 6  | VL fermion (1,2,-1/2) | VL lepton                           | 1/2   | 1/5             | 1/3             | 0               |
| 7  | VL fermion (3,2,1/6)  | VL quark                            | 1/2   | 1/15            | 1               | 1/3             |
| 8  | VL fermion (1,1,-1)   | VL e singlet                        | 1/2   | 2/5             | 0               | 0               |
| 9  | VL fermion (3,1,-1/3) | VL d singlet                        | 1/2   | 2/15            | 0               | 1/3             |
| 10 | VL fermion (3,1,2/3)  | VL u singlet                        | 1/2   | 8/15            | 0               | 1/3             |
| 11 | SU(5) 5+5 fermion     | Complete 5-plet                     | 1/2   | 2/5             | 1               | 1/3             |
| 12 | SU(5) 10+10 fermion   | Complete 10-plet                    | 1/2   | 6/5             | 1               | 1               |
| 13 | 2 × Scalar (1,2,1/2)  | Two Higgs                           | 0     | 1/5             | 1/3             | 0               |
| 14 | 3 × Scalar (1,2,1/2)  | Three Higgs                         | 0     | 3/10            | 1/2             | 0               |
| 15 | Full MSSM             | All partners                        | mixed | 5/2             | 25/6            | 4               |

## D.2: All Modified Gap Ratios and $M_{\text{GUT}}$ Values

| #  | Candidate                           | Modified<br>$b_1$   | Modified<br>$b_2$ | Modified<br>$b_3$ | Gap<br>Ratio             | Dist         | $\log_{10} M_{\text{GUT}}$ |
|----|-------------------------------------|---------------------|-------------------|-------------------|--------------------------|--------------|----------------------------|
| 1  | Scalar<br>(1,2,1/2)                 | 41/10+1/10-19/6+1/6 | -7                |                   | 9/5 =<br>1.800           | 0.442        | 13.9                       |
| 2  | Scalar<br>(3,1,-1/3)                | 41/10+1/15-19/6     | -7+1/6            |                   | 2.000                    | 0.642        | 13.7                       |
| 3  | Scalar<br>(3,2,1/6)                 | 41/10+1/30-19/6+1/2 | -7+1/6            |                   | 1.632                    | 0.274        | 14.6                       |
| 4  | Scalar<br>(1,3,0)                   | 41/10               | -19/6+1/3         | -7                | 1.664                    | 0.306        | 14.4                       |
| 5  | Scalar<br>(8,1,0)                   | 41/10               | -19/6             | -7+1/2            | 2.180                    | 0.822        | 13.8                       |
| 6  | VL<br>fermion<br>(1,2,-1/2)         | 41/10+1/5           | -19/6+1/3         | -7                | 1.712                    | 0.354        | 14.0                       |
| 7  | <b>VL<br/>fermion<br/>(3,2,1/6)</b> | <b>25/6</b>         | <b>-13/6</b>      | <b>-20/3</b>      | <b>38/27 =<br/>1.407</b> | <b>0.049</b> | <b>15.5</b>                |
| 8  | VL<br>fermion<br>(1,1,-1)           | 41/10+2/5           | -19/6             | -7                | 2.000                    | 0.642        | 13.2                       |
| 9  | VL<br>fermion<br>(3,1,-1/3)         | 41/10+2/15-19/6     | -7+1/3            |                   | 2.114                    | 0.756        | 13.6                       |
| 10 | VL<br>fermion<br>(3,1,2/3)          | 41/10+8/15-19/6     | -7+1/3            |                   | 2.229                    | 0.871        | 13.0                       |
| 11 | SU(5)<br>5+5                        | 9/2                 | -13/6             | -20/3             | 40/27 =<br>1.481         | 0.123        | 14.9                       |
| 12 | SU(5)<br>10+10                      | 41/10+6/5           | -19/6+1           | -7+1              | 1.948                    | 0.590        | 13.5                       |
| 13 | 2 $\times$<br>Scalar<br>(1,2,1/2)   | 41/10+1/5           | -19/6+1/3         | -7                | 1.712                    | 0.354        | 14.0                       |
| 14 | 3 $\times$<br>Scalar<br>(1,2,1/2)   | 41/10+3/10-19/6+1/2 | -7                |                   | 1.631                    | 0.273        | 14.1                       |
| 15 | <b>Full<br/>MSSM</b>                | <b>33/5</b>         | <b>1</b>          | <b>-3</b>         | <b>7/5 =<br/>1.400</b>   | <b>0.042</b> | <b>17.3</b>                |

## Appendix E: The Elimination Cascade

### E.1: Stage 1 — Gap Ratio Arithmetic

Threshold: modified gap ratio within 0.15 of measured 1.358.

Window: [1.208, 1.508].

| Candidate            | Gap Ratio | In window?       | Status     |
|----------------------|-----------|------------------|------------|
| Full MSSM            | 1.400     | Yes              | Survives   |
| VL quark doublet     | 1.407     | Yes              | Survives   |
| SU(5) 5+5            | 1.481     | Yes              | Survives   |
| 3 × Scalar doublet   | 1.631     | No (above 1.508) | Eliminated |
| Scalar leptoquark    | 1.632     | No               | Eliminated |
| Scalar SU(2) triplet | 1.664     | No               | Eliminated |
| VL lepton doublet    | 1.712     | No               | Eliminated |
| 2 × Scalar doublet   | 1.712     | No               | Eliminated |
| Scalar doublet       | 1.800     | No               | Eliminated |
| SU(5) 10+10̄         | 1.948     | No               | Eliminated |
| Scalar color triplet | 2.000     | No               | Eliminated |
| VL charged singlet   | 2.000     | No               | Eliminated |
| VL down singlet      | 2.114     | No               | Eliminated |
| Color octet scalar   | 2.180     | No               | Eliminated |
| VL up singlet        | 2.229     | No               | Eliminated |

12 eliminated. 3 survive.

### E.2: Stage 2 — Proton Decay

Bound:  $M_{\text{GUT}} > 10^{15.5} \text{ GeV}$  (Super-Kamiokande  $p \rightarrow e^+ \pi^0$ , model-dependent on GUT completion).

| Survivor         | M GUT       | Above bound?      | Status     |
|------------------|-------------|-------------------|------------|
| Full MSSM        | $10^{17.3}$ | Yes               | Survives   |
| VL quark doublet | $10^{15.5}$ | Yes (at boundary) | Survives   |
| SU(5) 5+5        | $10^{14.9}$ | No                | Eliminated |

1 eliminated. 2 survive.

### E.3: Stability Check

| Threshold    | Stage 1 survivors | After Stage 2    | Final survivors |
|--------------|-------------------|------------------|-----------------|
| 0.05 (tight) | MSSM, VL doublet  | MSSM, VL doublet | Same 2          |



| Threshold    | Stage 1 survivors           | After Stage 2    | Final survivors |
|--------------|-----------------------------|------------------|-----------------|
| 0.10         | MSSM, VL doublet, SU(5) 5+5 | MSSM, VL doublet | Same 2          |
| 0.15 (used)  | MSSM, VL doublet, SU(5) 5+5 | MSSM, VL doublet | Same 2          |
| 0.20 (loose) | MSSM, VL doublet, SU(5) 5+5 | MSSM, VL doublet | Same 2          |

The two survivors are the same regardless of the threshold choice between 0.05 and 0.20. The SU(5) 5+5 enters the window at threshold 0.13 but is always eliminated by proton decay at Stage 2.

## Appendix F: The Survivor's Properties

### F.1: Quantum Numbers

| Property               | Value                             |
|------------------------|-----------------------------------|
| SU(3) representation   | 3 (color triplet, fundamental)    |
| SU(2) representation   | 2 (weak doublet, fundamental)     |
| Hypercharge Y          | 1/6                               |
| Upper component charge | $Q = T_3 + Y = 1/2 + 1/6 = +2/3$  |
| Lower component charge | $Q = T_3 + Y = -1/2 + 1/6 = -1/3$ |
| SM analog              | Copy of (u L, d L) quark doublet  |
| Chirality              | Vector-like (L and R identical)   |
| Bare mass              | Allowed without Higgs mechanism   |
| Anomaly cancellation   | Automatic (vector-like)           |

### F.2: Beta Function Contributions

| $\Delta b_i$ | Value           | Fraction of SM $b_i$      | Effect on gap ratio         |
|--------------|-----------------|---------------------------|-----------------------------|
| $\Delta b_1$ | $1/15 = 0.0667$ | 1.6% of $b_1$             | Small increase in numerator |
| $\Delta b_2$ | $1 = 1.000$     | -31.6% of $b_2$ (reduces) | $b_2$                       |
| $\Delta b_3$ | $1/3 = 0.333$   | -4.8% of $b_3$ (reduces)  | $b_3$                       |

The dominant effect is  $\Delta b_2 = 1$ , which shifts  $b_2$  from  $-19/6 = -3.167$  to  $-13/6 = -2.167$ . This reduces  $b_1 - b_2$  (the gap ratio numerator) from  $109/15 = 7.267$  to  $19/3 = 6.333$ , a 13% decrease. The denominator  $b_2 - b_3$  increases from  $23/6 = 3.833$  to  $9/2 = 4.500$ , a 17% increase. Together these reduce the gap ratio from 1.896 to 1.407.

### F.3: The Asymmetry Ratio

| Candidate                    | $\Delta b_1$ | $\Delta b_2$ | $\Delta b_3$ | $\Delta b_2/\Delta b_1$ | Gap Ratio |
|------------------------------|--------------|--------------|--------------|-------------------------|-----------|
| VL quark doublet (3,2,1/6)   | 1/15         | 1            | 1/3          | 15.0                    | 1.407     |
| SU(5) 5+5                    | 2/5          | 1            | 1/3          | 2.5                     | 1.481     |
| VL lepton doublet (1,2,-1/2) | 1/5          | 1/3          | 0            | 1.67                    | 1.712     |
| VL charged singlet (1,1,-1)  | 2/5          | 0            | 0            | 0                       | 2.000     |
| Full MSSM                    | 5/2          | 25/6         | 4            | 1.67                    | 1.400     |

The VL quark doublet has the highest  $\Delta b_2/\Delta b_1$  ratio (15.0) of any single multiplet. This extreme asymmetry is why it works: the gap ratio numerator  $b_1 - b_2$  must shrink to fix the 40% overshoot, and adding  $\Delta b_2 \gg \Delta b_1$  does exactly that.

The MSSM achieves a similar gap ratio (1.400 vs 1.407) through a different mechanism: large contributions to ALL three betas that change the overall structure, not through asymmetry alone.

## Appendix G: The Running Equations

### G.1: One-Loop Running

$$1/\alpha_i(\mu) = 1/\alpha_i(M_Z) - b_i/(2\pi) \times \ln(\mu/M_Z)$$

### G.2: $M_{\text{GUT}}$ from $\alpha_1 = \alpha_2$ Crossing

$$1/\alpha_1(M_Z) - b_1 \times L = 1/\alpha_2(M_Z) - b_2 \times L, \text{ where } L = \ln(M_{\text{GUT}}/M_Z)/(2\pi)$$

$$L = (1/\alpha_1 - 1/\alpha_2)/(b_1 - b_2) = 31.5249 / (b_1 - b_2)$$

For each model,  $b_1 - b_2$  is the modified numerator:

| Model           | $b_1 - b_2$    | L     | $\ln(M_{\text{GUT}}/M_Z)$ | $\log_{10} M_{\text{GUT}}$ |
|-----------------|----------------|-------|---------------------------|----------------------------|
| SM              | 109/15 = 7.267 | 4.339 | 27.26                     | 13.80                      |
| SM + VL doublet | 19/3 = 6.333   | 4.977 | 31.27                     | 15.54                      |
| SM + SU(5) 5+5  | 20/3 = 6.667   | 4.729 | 29.71                     | 14.88                      |
| MSSM            | 28/5 = 5.600   | 5.630 | 35.37                     | 17.32                      |

### G.3: $\Delta(1/\alpha_3)$ at $M_{\text{GUT}}$

| Model           | $1/\alpha_3(M_{\text{GUT}})$ | $1/\alpha_1(M_{\text{GUT}})$ | $\Delta(1/\alpha_3)$ | Unification quality |
|-----------------|------------------------------|------------------------------|----------------------|---------------------|
| SM              | 38.84                        | 45.42                        | -6.58                | 14.5% miss          |
| SM + VL doublet | ~44.2                        | ~44.9                        | ~-0.7                | ~1.6% miss          |
| MSSM            | 25.36                        | 26.06                        | -0.69                | 2.7% miss           |

| Model | $1/\alpha_3(\text{M GUT})$ | $1/\alpha_1(\text{M GUT})$ | $\Delta(1/\alpha_3)$ | Unification quality |
|-------|----------------------------|----------------------------|----------------------|---------------------|
|-------|----------------------------|----------------------------|----------------------|---------------------|

## Appendix H: The Experimental Discriminator

### H.1: Proton Decay Predictions

| Scenario            | M GUT                   | Predicted $\tau(p \rightarrow e^+ \pi^0)$ | Hyper-K sensitivity          | Observable? |
|---------------------|-------------------------|-------------------------------------------|------------------------------|-------------|
| SM (no unification) | —                       | No prediction                             | —                            | No          |
| VL quark doublet    | $10^{15.5} \text{ GeV}$ | $\sim 10^{34-35} \text{ years}$           | $\sim 10^{35} \text{ years}$ | <b>Yes</b>  |
| Full MSSM           | $10^{17.3} \text{ GeV}$ | $\sim 10^{36-37} \text{ years}$           | $\sim 10^{35} \text{ years}$ | No          |

### H.2: The Decision Tree

Hyper-K observes proton decay at  $\tau \sim 10^{34-35} \text{ years} \rightarrow$  Consistent with VL doublet ( $M_{\text{GUT}} = 10^{15.5}$ ). Inconsistent with MSSM ( $M_{\text{GUT}} = 10^{17.3}$ ). Rules out SM (no unification).

Hyper-K sees nothing after full exposure ( $\sim 10^{35} \text{ year sensitivity}$ )  $\rightarrow$  Excludes VL doublet at minimal level. Consistent with MSSM or non-minimal extensions. Does not rule out unification (just rules out minimal single-particle fix).

### H.3: Other Experimental Handles

| Experiment       | Observable               | Current Status                   | VL Doublet Constraint            |
|------------------|--------------------------|----------------------------------|----------------------------------|
| LHC (CMS, ATLAS) | VL quark pair production | $M > \sim 1.5 \text{ TeV}$       | Mass bound, not representation   |
| HL-LHC           | Extended mass reach      | $\sim 2-3 \text{ TeV}$ projected | Tighter mass bound               |
| LEP/SLD legacy   | S, T parameters          | Precision data available         | Computable, not computed here    |
| Belle II, LHCb   | Flavor-changing currents | Ongoing                          | Model-dependent on Yukawa mixing |

## Appendix I: Scope and Bounds

### I.1: Why These Bounds

| Bound                       | Range            | Covers                                            | Excludes                                             |
|-----------------------------|------------------|---------------------------------------------------|------------------------------------------------------|
| $\dim(\text{SU}(3)) \leq 8$ | 1, 3, 3, 6, 6, 8 | All SU(5)/SO(10) components, all standard exotics | Higher reps (10, 15, ...)                            |
| $\dim(\text{SU}(2)) \leq 4$ | 1, 2, 3, 4<br>Y  | All standard GUT embeddings<br>$\leq 2$           | Higher multiplets (5, 7, ...) 0 to 2 in steps of 1/6 |
| Spin                        | 0, 1/2           | Scalars, VL fermions                              | Spin-1 (would modify gauge sector)                   |
| Count                       | 1 multiplet      | Minimal extensions                                | Multi-particle additions                             |

### I.2: Why Larger Representations Don't Help

Larger representations have larger Dynkin indices, producing larger beta function shifts. The SM gap ratio is already 40% above target. A successful candidate needs a SMALL, SPECIFIC correction — large  $\Delta b_2$  with small  $\Delta b_1$  — not a large overall shift. Examples:

| Hypothetical      | $\Delta b_3$ | Effect on gap ratio        | Outcome               |
|-------------------|--------------|----------------------------|-----------------------|
| (6,1,1/3) scalar  | 5/6          | Gap $\rightarrow \sim 2.5$ | Overshoots further    |
| (8,2,1/2) fermion | 2            | Gap $\rightarrow \sim 0.8$ | Undershoots massively |
| (3,3,1/3) fermion | 1            | Gap changes dramatically   | Outside target window |

The survivors are within the bounds. Nothing outside would survive.

## Appendix J: Verified Script Output

From the GUT running and BSM enumeration script (9/9 checks pass):

```
[PASS] Normalization:  $\sin^2 \theta_W$  from couplings
diff = 0.00e+00
```

[PASS] SM gap ratio = 218/115

$$1.8956521739$$

[PASS] MSSM gap ratio = 7/5

$$1.4000000000$$

[PASS] SM does not unify ( $\Delta > 5$ )

$$\Delta(1/\alpha_3) = -6.58$$

[PASS] MSSM nearly unifies ( $\Delta < 5$ )

$$\Delta(1/\alpha_3) = -0.69$$

[PASS]  $M_{\text{GUT}}(\text{SM}) > 10^{13}$

$$\log_{10} = 13.80$$

[PASS]  $M_{\text{GUT}}(\text{MSSM}) > 10^{16}$

$$\log_{10} = 17.32$$

[PASS] VL quark doublet gap < 0.05 from measured

$$\text{distance} = 0.049$$

[PASS] Measured gap ratio in [1.2, 1.5]

$$\text{gap} = 1.358193$$

All 15 candidates use the same verified infrastructure. Every gap ratio is computed in exact Fraction arithmetic from the same DATA-3 inputs. The MSSM gate confirms the beta function contributions are correctly encoded.

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## HOWL-PHYS-15: Integer-Forced Identification of the Minimal Unification Extension

This paper is the polished, self-contained presentation of the BSM enumeration result. Where PHYS-13 introduced the gap ratio and found the VL quark doublet, and PHYS-14 showed why fermions cancel, PHYS-15 presents the complete argument from measured couplings to experimental test in one clean logical chain.

**The method is constraint-driven enumeration, not model selection.** Start with two

inputs: the measured gap ratio (1.358) and the exact rational beta coefficients. Compute the mismatch. Enumerate every single-particle extension within bounded representations. Compute each modified gap ratio as an exact rational. Eliminate by arithmetic and existing data.

**The scope is explicit:** single new multiplet,  $SU(3) \leq 8$ ,  $SU(2) \leq 4$ ,  $|Y| \leq 2$ , scalar or vector-like fermion. These bounds cover all standard GUT representations plus common exotics. Larger representations produce larger beta shifts, which push further from target — the search is conservative.

**The elimination cascade:**

Stage 1 (arithmetic): 12 of 15 candidates have gap ratios more than 0.15 from measured. Window [1.208, 1.508]. Three survive: MSSM ( $7/5 = 1.400$ ), VL doublet ( $38/27 = 1.407$ ),  $SU(5)$  5+5 ( $40/27 = 1.481$ ).

Stage 2 (proton decay): Super-K bound  $M_{GUT} > 10^{15.5}$ . The  $SU(5)$  5+5 has  $M_{GUT} = 10^{14.9}$  — eliminated. Two survive.

Stage 3 (stability): Tightening the threshold from 0.15 to 0.05 eliminates the 5+5 earlier but leaves the same two survivors. Loosening to 0.20 admits the 5+5 into Stage 2 where it's still killed by proton decay. Result is stable.

**The VL quark doublet (3,2,1/6) works because of its extreme asymmetry:**  $\Delta b_2/\Delta b_1 = 15$ . No other single multiplet has this ratio. The small hypercharge  $Y = 1/6$  (smallest among all colored weak-doublet candidates) minimizes  $\Delta b_1$  while the (3,2) representation maximizes  $\Delta b_2$ . The quantum numbers of the left-handed quark doublet are special for unification — not by choice, but by arithmetic.

**The experimental discriminator is clean:** Hyper-K sees proton decay at  $10^{34-35}$  years → consistent with VL doublet, inconsistent with MSSM. Hyper-K sees nothing → VL doublet excluded at minimal level, MSSM or non-minimal extensions survive.

**What distinguishes this from the standard literature:** The direction is inverted. Standard approach: choose a model → compute predictions → compare. This analysis: measure the mismatch → enumerate all possibilities → eliminate by arithmetic. The result identifies a known particle type — VL quarks are well-studied. What's new is that the identification comes from elimination, not preference.

The complete physics stack:

- **PHYS-1-4:** Foundations, boundaries, test program
- **PHYS-5-6:**  $\alpha$  running, confinement two-face
- **PHYS-7:**  $\theta = 0$  (19→18)
- **PHYS-8:** Koide (18→17 conditional)
- **PHYS-9:**  $\alpha$  from  $a_e$ , 4.3 ppb
- **PHYS-10-11:** Remainder framework, nine domains,  $R_2$  universal
- **PHYS-12:** Electroweak integer anatomy,  $A_2$  decomposition

- **PHYS-13:** Gap ratio 218/115, BSM enumeration (first pass)
- **PHYS-14:** Unified map, fermion cancellation theorem
- **PHYS-15:** Complete constraint-driven enumeration, two survivors, experimental test

Aligned. Ready for the appendix tables.

## APPENDIX K: THE LOGICAL CHAIN — EVERY STEP WITH ITS TYPE

| Step | Input                                                        | Operation                                             | Output                                                                         | Type                           | Reversible?                            |
|------|--------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------|----------------------------------------|
| 1    | $SU(3) \times SU(2) \times U(1)$ gauge group                 | Representation theory (Casimirs, Dynkin indices)      | $b_1^{\text{gauge}} = 0, b_2^{\text{gauge}} = -22/3, b_3^{\text{gauge}} = -11$ | Mathematical theorem           | No — follows from the gauge group      |
| 2    | SM fermion content (3 generations $\times 5+10$ of $SU(5)$ ) | Dynkin index sum                                      | Per-gen $(\Delta b_1, \Delta b_2, \Delta b_3) = (4/3, 4/3, 4/3)$               | Mathematical theorem           | No — follows from anomaly cancellation |
| 3    | SM Higgs (1,2,1/2)                                           | Complex scalar beta formula                           | $(\Delta b_1, \Delta b_2, \Delta b_3) = (1/10, 1/6, 0)$                        | Mathematical theorem           | No — follows from the representation   |
| 4    | Steps 1+2+3                                                  | Addition of exact rationals                           | $b_1 = 41/10, b_2 = -19/6, b_3 = -7$                                           | Exact arithmetic               | Verified (Gate 1)                      |
| 5    | Step 4                                                       | $(b_1 - b_2)/(b_2 - b_3)$                             | Gap ratio = $218/115 = 1.896$                                                  | Exact arithmetic               | Verified                               |
| 6    | $\alpha_{\text{em}}, \sin^2\theta_W, \alpha_s$ (DATA-3)      | GUT normalization formulas                            | $1/\alpha_1 = 63.210, 1/\alpha_2 = 31.685, 1/\alpha_3 = 8.475$                 | Measurement + exact conversion | Verified ( $\sin^2\theta_W$ roundtrip) |
| 7    | Step 6                                                       | $(1/\alpha_1 - 1/\alpha_2)/(1/\alpha_2 - 1/\alpha_3)$ | Measured gap ratio = 1.358                                                     | Measured ratio                 | —                                      |

| Step | Input                                         | Operation                 | Output                                              | Type                          | Reversible?             |
|------|-----------------------------------------------|---------------------------|-----------------------------------------------------|-------------------------------|-------------------------|
| 8    | Steps 5 and 7                                 | Comparison                | 218/115 $\neq$ 1.358 (40% miss)                     | Arithmetic                    | The unification failure |
| 9    | Scope bounds (SU(3) $\leq$ 8, SU(2) $\leq$ 4, | Y                         | $\leq$ 2, spin 0 or 1/2)                            | Enumerate all representations | 15 candidate multiplets |
| 10   | Each candidate's quantum numbers              | Dynkin index formulas     | 15 sets of ( $\Delta b_1, \Delta b_2, \Delta b_3$ ) | Mathematical theorem          | Each is exact           |
| 11   | SM betas + each $\Delta b_i$                  | Rational arithmetic       | 15 modified gap ratios (exact)                      | Exact arithmetic              | Each verified           |
| 12   | Modified gap ratios vs 1.358                  | Distance < 0.15 threshold | 3 survive Stage 1                                   | Criterion (stated)            | Stable under variation  |
| 13   | M GUT for survivors                           | Running equation          | 3 unification scales                                | Exact arithmetic              | Verified                |
| 14   | M GUT > $10^{15.5}$ (Super-K bound)           | Comparison                | 1 eliminated (SU(5) 5+5)                            | Experimental data             | Model-dependent         |
| 15   | Two survivors                                 | Statement                 | MSSM (7/5) and VL doublet (38/27)                   | Result                        | —                       |

**Every step is either a mathematical theorem, exact arithmetic, a measurement, or a stated criterion.** No step involves preference, aesthetic judgment, or model selection. The result is forced by the integers and the data.

## APPENDIX L: WHY $Y = 1/6$ IS THE WINNING HYPERCHARGE

The VL quark doublet's small hypercharge is the key to its extreme asymmetry. This table shows what happens for all possible hypercharges of a color-triplet weak-doublet.



| Representation |       | Y     |   | $\Delta b_1$ | $\Delta b_2$ | $\Delta b_3$ | $\Delta b_2/\Delta b_1$ | Gap Ratio |
|----------------|-------|-------|---|--------------|--------------|--------------|-------------------------|-----------|
| (3,2,1/6) 1/6  | 0.167 | 1/15  | 1 | 1/3          | <b>15.0</b>  | <b>38/27</b> | <b>0.049</b>            |           |
|                |       |       |   |              |              | =            |                         |           |
|                |       |       |   |              |              | <b>1.407</b> |                         |           |
| (3,2,1/2) 1/2  | 0.500 | 1/3   | 1 | 1/3          | 3.0          | ~1.53        | 0.17                    |           |
| (3,2,5/6) 5/6  | 0.833 | 5/6   | 1 | 1/3          | 1.2          | ~1.66        | 0.30                    |           |
| (3,2,7/6) 7/6  | 1.167 | 49/30 | 1 | 1/3          | 0.61         | ~1.78        | 0.42                    |           |
| (3,2,-1/6) 1/6 | 0.167 | 1/15  | 1 | 1/3          | 15.0         | 38/27        | 0.049                   |           |
|                |       |       |   |              |              | =            |                         |           |
|                |       |       |   |              |              | 1.407        |                         |           |

| Representation                        | Y                                                                                                                                                                                                                                                                                                                   | $\Delta b_1$ | $\Delta b_2$ | $\Delta b_3$ | Gap<br>$\Delta b_2/\Delta b_1$ Ratio |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|--------------|--------------------------------------|
| $\Delta b_1 \propto Y^2$ .<br>Smaller | Y means smaller $\Delta b_1$ , $\Delta b_2$ and $\Delta b_3$ are independent of Y (they depend only on the SU(2) and SU(3) representations). The asymmetry $\Delta b_2/\Delta b_1$ therefore scales as $1/Y^2$ . At $Y = 1/6$ (the smallest hypercharge consistent with integer electric charges in a doublet), the |              |              |              |                                      |

**The quantum numbers (3,2,1/6) are the quantum numbers of the SM left-handed quark doublet.** This is not a coincidence — it is the representation with the smallest hypercharge among all colored weak doublets, and therefore the one with the highest  $\Delta b_2/\Delta b_1$  ratio, and therefore the one closest to fixing the gap ratio with a single particle.

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## APPENDIX M: THE TWO SURVIVORS — SIDE BY SIDE

| Property                                        | VL Quark Doublet<br>(3,2,1/6)                                 | Full MSSM                                      |
|-------------------------------------------------|---------------------------------------------------------------|------------------------------------------------|
| <b>Particle content</b>                         | 1 new multiplet (2 Weyl fermions: $Q = +2/3$ and $Q = -1/3$ ) | ~30 new multiplets (~120 new fields)           |
| <b>Gap ratio</b>                                | $38/27 = 1.40741$                                             | $7/5 = 1.40000$                                |
| <b>Distance from measured 1.358</b>             | 0.049                                                         | 0.042                                          |
| <b>M GUT</b>                                    | $10^{15.5}$ GeV                                               | $10^{17.3}$ GeV                                |
| <b><math>\Delta(1/\alpha_3)</math> at M GUT</b> | ~-0.7                                                         | -0.69                                          |
| <b>Unification quality</b>                      | ~1.6% miss                                                    | 2.7% miss                                      |
| <b>Proton lifetime</b>                          | ~ $10^{34-35}$ years                                          | ~ $10^{36-37}$ years                           |
| <b>Hyper-K testable?</b>                        | <b>Yes</b> — within projected sensitivity                     | No — beyond projected sensitivity              |
| <b>Dark matter candidate?</b>                   | No                                                            | Yes (neutralino)                               |
| <b>Hierarchy stabilization?</b>                 | No                                                            | Yes (cancellation of quadratic divergences)    |
| <b>Anomaly cancellation</b>                     | Automatic (vector-like)                                       | Automatic (SUSY)                               |
| <b>LHC direct search bound</b>                  | M VL > ~1.5 TeV                                               | No SUSY partners found through Run 3           |
| <b>Gap ratio numerator and denominator</b>      | 38 and 27 (two-digit integers)                                | 7 and 5 (single-digit integers)                |
| <b>Gap ratio factored</b>                       | $38 = 2 \times 19$ , $27 = 3^3$                               | 7 (prime), 5 (prime)                           |
| <b>Number of free parameters added</b>          | 1 (mass of VL doublet)                                        | ~105 (soft SUSY breaking parameters)           |
| <b>Mechanism</b>                                | Asymmetric beta shift: $\Delta b_2/\Delta b_1 = 15$           | Balanced restructuring of all three betas      |
| <b>What it solves</b>                           | Gauge coupling unification only                               | Unification + hierarchy + dark matter + flavor |
| <b>What it doesn't solve</b>                    | Hierarchy, dark matter, strong CP, flavor                     | Strong CP (without axion), flavor hierarchy    |

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## APPENDIX N: THE ELIMINATION CASCADE — VISUAL SUMMARY

Starting with 15 candidates and progressively eliminating.

| Candidate               | Gap Ratio                 | Stage 1 (< 0.15<br>from 1.358)? | Stage 2 (M<br>GUT ><br>$10^{15.5}$ )? | Final Status          |
|-------------------------|---------------------------|---------------------------------|---------------------------------------|-----------------------|
| Full MSSM               | $7/5 = 1.400$             | <b>PASS</b> (dist<br>0.042)     | <b>PASS</b> ( $10^{17.3}$ )           | <b>SURVIVOR</b>       |
| VL (3,2,1/6)            | $38/27 = 1.407$           | <b>PASS</b> (dist<br>0.049)     | <b>PASS</b> ( $10^{15.5}$ )           | <b>SURVIVOR</b>       |
| SU(5) 5+5               | $40/27 = 1.481$           | <b>PASS</b> (dist<br>0.123)     | <b>FAIL</b> ( $10^{14.9}$ )           | Eliminated<br>Stage 2 |
| 3 × Scalar<br>(1,2,1/2) | $106/65 \approx$<br>1.631 | FAIL (dist<br>0.273)            | —                                     | Eliminated<br>Stage 1 |
| Scalar<br>(3,2,1/6)     | $\approx 1.632$           | FAIL (dist<br>0.274)            | —                                     | Eliminated<br>Stage 1 |
| Scalar (1,3,0)          | $\approx 1.664$           | FAIL (dist<br>0.306)            | —                                     | Eliminated<br>Stage 1 |
| VL (1,2,-1/2)           | $\approx 1.712$           | FAIL (dist<br>0.354)            | —                                     | Eliminated<br>Stage 1 |
| 2 × Scalar<br>(1,2,1/2) | $\approx 1.712$           | FAIL (dist<br>0.354)            | —                                     | Eliminated<br>Stage 1 |
| Scalar<br>(1,2,1/2)     | $9/5 = 1.800$             | FAIL (dist<br>0.442)            | —                                     | Eliminated<br>Stage 1 |
| SU(5) 10+10             | $\approx 1.948$           | FAIL (dist<br>0.590)            | —                                     | Eliminated<br>Stage 1 |
| Scalar<br>(3,1,-1/3)    | 2.000                     | FAIL (dist<br>0.642)            | —                                     | Eliminated<br>Stage 1 |
| VL (1,1,-1)             | 2.000                     | FAIL (dist<br>0.642)            | —                                     | Eliminated<br>Stage 1 |
| VL (3,1,-1/3)           | $\approx 2.114$           | FAIL (dist<br>0.756)            | —                                     | Eliminated<br>Stage 1 |
| Scalar (8,1,0)          | $\approx 2.180$           | FAIL (dist<br>0.822)            | —                                     | Eliminated<br>Stage 1 |
| VL (3,1,2/3)            | $\approx 2.229$           | FAIL (dist<br>0.871)            | —                                     | Eliminated<br>Stage 1 |

**15 → 3 (Stage 1) → 2 (Stage 2).** The gap between rank 2 and rank 3 is a factor of 2.5 in distance (0.049 vs 0.123). The gap between rank 3 and rank 4 is a factor of 2.2 (0.123 vs 0.273). The two survivors are well separated from everything else.

## APPENDIX O: THE GAP RATIO RATIONALS — ALGEBRAIC STRUCTURE

Every exact gap ratio from the enumeration, showing the factorizations.

| Candidate  | Gap Ratio<br>(exact) | Numerator<br>Factored | Denominator<br>Factored | Digit Sum<br>N+D | Complexity<br>Rank |
|------------|----------------------|-----------------------|-------------------------|------------------|--------------------|
| MSSM       | <b>7/5</b>           | 7                     | 5                       | 12               | 1 (simplest)       |
| VL         | <b>38/27</b>         | $2 \times 19$         | $3^3$                   | 65               | 2                  |
| (3,2,1/6)  |                      |                       |                         |                  |                    |
| SU(5) 5+5  | <b>40/27</b>         | $2^3 \times 5$        | $3^3$                   | 67               | 3                  |
| SM         | <b>218/115</b>       | $2 \times 109$        | $5 \times 23$           | 333              | 4                  |
| Scalar     | <b>9/5</b>           | $3^2$                 | 5                       | 14               | —                  |
| (1,2,1/2)  |                      |                       |                         |                  |                    |
| Scalar     | <b>2/1</b>           | 2                     | 1                       | 3                | —                  |
| (3,1,-1/3) |                      |                       |                         |                  |                    |
| VL         | <b>2/1</b>           | 2                     | 1                       | 3                | —                  |
| (1,1,-1)   |                      |                       |                         |                  |                    |
| Scalar     | <b>109/50</b>        | 109                   | $2 \times 5^2$          | 159              | —                  |
| (8,1,0)    |                      |                       |                         |                  |                    |

### Observations:

The two survivors have the simplest gap ratios among candidates close to the target: 7/5 (single-digit integers) and 38/27 (two-digit integers). The SM's 218/115 has three-digit integers. Algebraic simplicity correlates with proximity to unification.

The prime 109 appears in both the SM gap ratio ( $218 = 2 \times 109$ ) and the scalar octet gap ratio (109/50). This integer comes from the specific combination of gauge self-couplings and Higgs contribution:  $b_1 - b_2 = 109/15$ . Any candidate that doesn't change  $b_1 - b_2$  retains the factor 109.

The VL doublet's gap ratio 38/27 shares no prime factors with the SM's 218/115. The doublet fundamentally restructures the numerator (from 109/15 to 19/3) and denominator (from 23/6 to 9/2). This is a complete replacement, not a small correction.

## APPENDIX P: THE EXPERIMENTAL DECISION TREE — DETAILED

| Observation                                                       | Interpretation for VL Doublet               | Interpretation for MSSM                                   | Next Step                         |
|-------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------|-----------------------------------|
| Hyper-K sees $p \rightarrow e^+ \pi^0$ at $\tau \sim 10^{34}$ yr  | <b>Consistent</b> (M GUT $\sim 10^{15.5}$ ) | Excluded at this scale (M GUT too high for this lifetime) | Search for VL quarks at LHC       |
| Hyper-K sees $p \rightarrow e^+ \pi^0$ at $\tau \sim 10^{35}$ yr  | Consistent (upper range of prediction)      | Excluded                                                  | Same                              |
| Hyper-K sees nothing after 10 yr ( $\sim 10^{35}$ yr sensitivity) | <b>Excluded</b> at minimal SU(5) level      | Consistent ( $\tau \sim 10^{36-37}$ yr, beyond reach)     | Look for SUSY at future colliders |
| LHC finds VL quarks at $\sim 2$ TeV                               | Mass determined; check EW precision         | Not directly relevant to MSSM                             | Compute S, T; refine M GUT        |
| LHC finds VL quarks at $\sim 5$ TeV (HL-LHC)                      | Mass determined; still consistent           | Not relevant                                              | Same                              |
| LHC finds no VL quarks up to 3 TeV                                | Mass $> 3$ TeV; still viable                | Not relevant                                              | Wait for higher energy collider   |
| LHC finds SUSY partners                                           | VL doublet may coexist with MSSM            | <b>Confirmed</b> as particle physics                      | Compute combined gap ratio        |
| EW precision (S, T) excludes VL doublet                           | <b>Excluded</b> by indirect data            | Not affected                                              | MSSM or other solutions           |

**The cleanest discriminator is proton decay.** The VL doublet and MSSM predict different proton lifetimes by a factor of 100-1000 ( $10^{34-35}$  vs  $10^{36-37}$  years). Hyper-K's projected sensitivity of  $\sim 10^{35}$  years falls between these predictions, making it the ideal experiment.

## APPENDIX Q: THE SERIES PARAMETER REDUCTION — FINAL SCORECARD

| # | Parameter    | Original Status | Current Status     | Paper  | Method                                        | Conditional? |
|---|--------------|-----------------|--------------------|--------|-----------------------------------------------|--------------|
| 1 | $\theta$ QCD | Free (1 of 19)  | <b>Derived = 0</b> | PHYS-7 | Ground state of $\mathbb{Z}$ -periodic energy | No           |

| #            | Parameter                         | Original Status                 | Current Status                              | Paper      | Method                                                 | Conditional?            |
|--------------|-----------------------------------|---------------------------------|---------------------------------------------|------------|--------------------------------------------------------|-------------------------|
| 2            | $m_\tau$                          | Free (Yukawa)                   | <b>Derived from <math>m_e, m_\mu</math></b> | PHYS-8     | Koide: $(1+a^2/2)/N = 2/3$ at $N=3, a^2=2$             | Yes $(0.91\sigma)$      |
| 3            | $\alpha^{-1} \leftrightarrow a_e$ | Free (one of the two)           | <b>Relabeled</b> (not reduced)              | PHYS-9     | QED series inversion                                   | N/A                     |
| 4            | $\sin^2\theta_W$                  | Free                            | <b>Path identified, blocked</b>             | PHYS-13-15 | Gap ratio $\leftrightarrow \sin^2\theta_W$ equivalence | Blocked by BSM spectrum |
| 5-19         | 15 remaining                      | Free                            | Free                                        | —          | No known law                                           | —                       |
| <b>Count</b> | <b>19 free</b>                    | <b>17 (or 18 unconditional)</b> |                                             |            |                                                        |                         |

**What PHYS-15 adds to the scorecard:** Not a parameter reduction, but a structural result — the gap ratio mismatch 218/115 vs 1.358 is diagnosed (it’s a boson problem, not a fermion problem), and the space of single-particle solutions is mapped (two survivors: MSSM and VL doublet). If the VL doublet is confirmed,  $\sin^2\theta_W$  could potentially move from “blocked” to “derived,” conditional on the GUT completion.

## APPENDIX R: THE COMPLETE STACK — EVERY PAPER’S CONTRIBUTION TO PHYS-15

| Paper  | What PHYS-15 Uses From It                          | How It Enters                                         |
|--------|----------------------------------------------------|-------------------------------------------------------|
| PHYS-1 | Mass = inertia; soliton boundary framework         | Each threshold is a PHYS-1 boundary                   |
| PHYS-2 | Transformation laws are integers; couplings run    | The beta coefficients ARE the transformation law      |
| PHYS-5 | $\alpha_{em}$ running in exact Fraction arithmetic | Infrastructure for the EM sector of the running       |
| PHYS-6 | Confinement two-face structure                     | The blank zone in the map ( $\sim 0.3\text{--}2$ GeV) |
| PHYS-7 | $\theta_{QCD} = 0$ (19→18 parameters)              | The parameter count starts at 18, not 19              |

| Paper   | What PHYS-15 Uses From It                                 | How It Enters                                                 |
|---------|-----------------------------------------------------------|---------------------------------------------------------------|
| PHYS-8  | $m_\tau$ from Koide (18 $\rightarrow$ 17 conditional)     | The parameter count is 17 conditional                         |
| PHYS-9  | $\alpha$ from a e; QED series is integers                 | Demonstrates the integer structure at one coupling            |
| PHYS-10 | Remainder framework; PSLQ null on SM constants            | The gap ratio mismatch is not resolved by basis constants     |
| PHYS-11 | $R_2$ universal; three subgroups                          | The $2\pi$ in the running equation is $8R_2$                  |
| PHYS-12 | Electroweak integer anatomy; 14/14 checks                 | Infrastructure: G F, M Z, $\sin^2\theta_W$ as exact Fractions |
| PHYS-13 | Gap ratio 218/115; BSM enumeration; VL doublet identified | The enumeration and finding — PHYS-15 is the polished version |
| PHYS-14 | Fermion cancellation theorem; unified map                 | The structural insight: gap ratio is a boson problem          |

**PHYS-15 is the synthesis.** Every prior paper contributes a piece. The Fraction arithmetic infrastructure (MATH-2, PHYS-5, PHYS-9, PHYS-12) ensures exact computation. The structural insights (PHYS-14 fermion cancellation, PHYS-6 confinement wall) constrain the analysis. The DATA-3 measurements provide the target. The enumeration (PHYS-13) provides the method. PHYS-15 assembles these into one self-contained argument from measured couplings to experimental test.

[[HOWL-PHYS-15-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-15-2026>

[[HOWL-PHYS-1-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-1-2026>

[[HOWL-PHYS-2-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-2-2026>

[[HOWL-PHYS-6-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-6-2026>

[[HOWL-PHYS-7-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-7-2026>

[[HOWL-PHYS-8-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-8-2026>



[[HOWL-PHYS-9-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-9-2026>

[[HOWL-PHYS-10-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-10-2026>

[[HOWL-PHYS-11-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-11-2026>

[[HOWL-PHYS-12-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-12-2026>

[[HOWL-PHYS-13-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026>